

BCP for Continuous-Trace C^* -Algebras with Separable Fibres

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K , has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

We use the standard Dixmier–Douady/Fell local description of continuous-trace C^* -algebras: if A is continuous trace and $X = \text{Prim}(A)$, then X is locally compact Hausdorff, A is a continuous $C_0(X)$ -algebra, and locally A is the algebra of C_0 -sections of an elementary C^* -bundle with fibre $\mathcal{K}(H)$. Equivalently, for every $x \in X$ there is an open neighborhood U and a Hilbert space H_U such that the ideal A_U with primitive spectrum U is $C_0(U)$ -isomorphic to

$$C_0(U, \mathcal{K}(H_U)).$$

Theorem 1. *Let A be a continuous-trace C^* -algebra. Put $X = \text{Prim}(A)$. Assume that all local elementary fibres are compact operators on separable nonzero Hilbert spaces. Then the following are equivalent:*

1. A has BCP;

2. A has UBCP;
3. X has a countable π -basis.

Thus, within the continuous-trace/separable-fibre class, the exact topological criterion is the same countable π -basis condition as in the commutative case.

Idea of the Proof

The negative implication is the primitive-space obstruction: every C^* -algebra with BCP has a primitive ideal space with countable π -basis.

For the positive implication, countable generation of the algebra is not needed. The local elementary-bundle structure gives charts $A_U \simeq C_0(U, \mathcal{K}(H))$, and separability of H makes each local $\mathcal{K}(H)$ norm separable. A countable π -basis lets us choose countably many charted open patches cofinal among all nonempty open sets, and inside each such chart countably many compactly supported scalar bumps. The covering centers are simply

$$4hT,$$

where h is one of those local bumps and T is one of the countably many unit compact operators in the local chart. Given $a \in S_A$, choose a charted patch where $\|a(x)\| > 3/4$, choose T close to the local fibre direction of a , and then choose the bump small enough that the resulting center remains within distance $7/2$ of a .

Proof

We first record the negative half, proved in the companion primitive-space packet [4].

Lemma 1 (primitive π -base obstruction). *If a C^* -algebra B has BCP, then $\text{Prim}(B)$ has a countable π -basis.*

It remains to prove UBCP when X has a countable π -basis.

Lemma 2 (charted bump family). *Assume X has a countable π -basis and A is as in Theorem 1. There are countably many triples*

$$(O_n, \Phi_n, \mathcal{H}_n)_{n \geq 1}$$

with the following properties:

1. each $O_n \subset X$ is nonempty open;
2. $\Phi_n : A_{O_n} \rightarrow C_0(O_n, \mathcal{K}(\mathcal{H}_n))$ is a $C_0(O_n)$ -isomorphism, with $\mathcal{H}_n$ separable and nonzero;
3. the family (O_n) is a π -basis of X .

Moreover, for each n there is a countable family of scalar functions $\mathcal{B}_n \subset C_c(X)$ such that $0 \leq h \leq 1$, $\|h\|_\infty = 1$, $\text{supp } h \subset O_n$, and for every nonempty open $W \subset O_n$ some $h \in \mathcal{B}_n$ satisfies $\text{supp } h \subset W$.

Proof. Let $(V_j)_{j \geq 1}$ be a countable π -basis for X . For each j , choose $x_j \in V_j$. By local triviality, choose an open neighborhood T_j of x_j and a $C_0(T_j)$ -isomorphism

$$A_{T_j} \cong C_0(T_j, \mathcal{K}(H_j)).$$

Set $O_j = V_j \cap T_j$, and restrict the isomorphism to O_j . Then (O_j) is still a π -basis: if $W \subset X$ is nonempty open, choose $V_j \subset W$; then O_j is nonempty and $O_j \subset W$.

Fix n . For every j with $V_j \subset O_n$, local compactness and regularity give a function $h_{n,j} \in C_c(X)$ such that

$$0 \leq h_{n,j} \leq 1, \quad \|h_{n,j}\|_\infty = 1, \quad \text{supp } h_{n,j} \subset V_j.$$

Let $\mathcal{B}_n$ be the countable family of all such $h_{n,j}$. If $W \subset O_n$ is nonempty open, choose $V_j \subset W$. Then $V_j \subset O_n$, and $h_{n,j} \in \mathcal{B}_n$ has support contained in W . $\square$

Proof of Theorem 1. UBCP implies BCP by definition, and Lemma 1 gives $\text{BCP} \Rightarrow$ countable π -basis of X . We prove the converse.

Assume X has a countable π -basis. Let $(O_n, \Phi_n, \mathcal{H}_n)$ and $\mathcal{B}_n$ be as in Lemma 2. Since each $\mathcal{H}_n$ is separable, $\mathcal{K}(\mathcal{H}_n)$ is separable. Choose a countable $1/64$ -net

$$D_n \subset S_{\mathcal{K}(\mathcal{H}_n)}$$

for the unit sphere of $\mathcal{K}(\mathcal{H}_n)$.

For each n , $h \in \mathcal{B}_n$, and $T \in D_n$, define

$$c(n, h, T) \in A$$

to be the element whose image under Φ_n is the compactly supported section $x \mapsto 4h(x)T$ on O_n . This element belongs to A_{O_n} , viewed as an ideal of A . The family of all such centers is countable and

$$\|c(n, h, T)\| = 4,$$

because $\|h\|_\infty = 1$ and $\|T\| = 1$. Hence every ball

$$B(c(n, h, T), 7/2)$$

misses the open ball $B(0, 1/2)$.

We claim that these balls cover S_A . Let $a \in S_A$. Choose a nonempty open set

$$W_0 = \{x \in X : \|a(x)\| > 3/4\}.$$

Since (O_n) is a π -basis, choose n with $O_n \subset W_0$, and choose $y \in O_n$.

Choose $\chi \in C_c(X)$ with $0 \leq \chi \leq 1$, $\text{supp } \chi \subset O_n$, and $\chi = 1$ on a neighborhood of y . Put

$$f = \Phi_n(\chi a) \in C_0(O_n, \mathcal{K}(\mathcal{H}_n)).$$

At y , this is the local coordinate representative of $a(y)$. Let

$$\alpha = \|f(y)\| = \|a(y)\| > 3/4, \quad u = f(y)/\alpha.$$

Pick $T \in D_n$ with $\|T - u\| < 1/64$.

For $0 \leq s \leq 1$, the map $s \mapsto \|\alpha u - 4sT\|$ is convex. At $s = 0$ its value is $\alpha \leq 1$. At $s = 1$,

$$\|\alpha u - 4T\| \leq \alpha\|u - T\| + (4 - \alpha)\|T\| < 4 - \alpha(1 - 1/64) < 7/2.$$

Therefore

$$\sup_{0 \leq s \leq 1} \|f(y) - 4sT\| < 7/2.$$

By continuity of f , after shrinking around y inside the set where $\chi = 1$, there is a nonempty open neighborhood $N \subset O_n$ of y such that

$$\sup_{0 \leq s \leq 1} \|f(x) - 4sT\| < 7/2 \quad (x \in N).$$

Use Lemma 2 to choose $h \in \mathcal{B}_n$ with $\text{supp } h \subset N$. Let $c = c(n, h, T)$.

If $x \in \text{supp } h$, then $\chi(x) = 1$, so $f(x)$ is the local coordinate representative of $a(x)$. With $s = h(x)$, the preceding estimate gives

$$\|a(x) - c(x)\| < 7/2.$$

If $x \notin \text{supp } h$, then $c(x) = 0$, hence

$$\|a(x) - c(x)\| = \|a(x)\| \leq 1 < 7/2.$$

Taking the supremum over $x \in X$, we obtain $\|a - c\| < 7/2$. Thus a is covered by one of the countably many balls $B(c(n, h, T), 7/2)$.

The countable family of radius $7/2$ balls covers S_A and uniformly avoids a neighborhood of the origin. Hence A has UBCP. $\square$

Scope and Limitations

This is a positive classification theorem for continuous-trace C^* -algebras with separable elementary fibres. It strictly extends the previous countably generated continuous $C_0(X)$ -algebra packet: countable generation over $C_0(X)$ is replaced by local elementary triviality plus separable fibre models.

The theorem is not yet a full arbitrary C^* -algebra characterization. Earlier packets show why such a characterization cannot be purely primitive-topological: a II_1 factor has one-point primitive ideal space but fails BCP by the atomless von Neumann theorem of the source paper. The present result says that, on the type-I/continuous-trace side, the commutative π -basis criterion does survive exactly.

Novelty and Search Bounds

The run indexes were searched for arXiv:2311.14257, continuous trace, C^* -algebras, BCP, UBCP, $C_0(X)$ -algebras, local fibre nets, and π -bases. Web searches on June 17, 2026 for exact phrases such as *ball-covering property continuous trace C^* -algebra* and *BCP continuous trace C^* -algebra π -base* found no exact existing statement of this theorem. The proof uses standard continuous-trace local triviality from the Dixmier–Douady/Fell theory and the primitive π -base obstruction proved in the companion packet.

Human Review Focus

The main checks are:

- the exact convention for continuous-trace local triviality used in the statement;
- the charted bump Lemma 2, especially the fact that a countable π -basis can be refined to countably many charted patches;

- the cutoff step χa , which ensures the local representative f is a genuine $C_0(O_n, \mathcal{K}(\mathcal{H}_n))$ -section even though a itself need not restrict to an element of the local ideal;
- the fixed constants $1/64$, $7/2$, and center norm 4 .

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Jacques Dixmier, *C^* -algebras*, North-Holland, 1977.
- [3] Iain Raeburn and Dana P. Williams, *Morita Equivalence and Continuous-Trace C^* -Algebras*, American Mathematical Society, 1998.
- [4] Run `fa_banach_001`, lane 19, *A Primitive-Space π -Base Obstruction for C^* -Algebras with BCP*, solution packet `2311.14257_primitive_pi_base_obstruction`, 2026.
- [5] Run `fa_banach_001`, lane 19, *BCP for Countably Generated Continuous $C_0(X)$ -Algebras*, solution packet `2311.14257_countably_generated_continuous_c0x_bcp`, 2026.